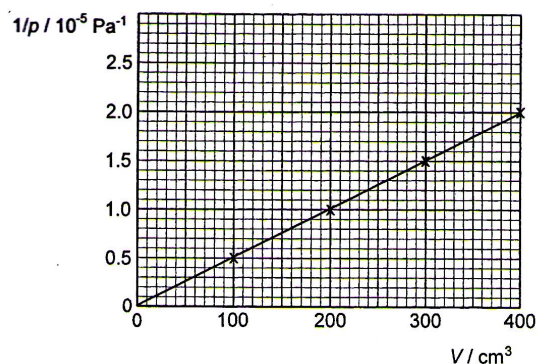


- 1 (a) (i) A graph of $\frac{1}{p}$ is plotted against V .

V / m^3	$p / 10^5 \text{ Pa}$	$\frac{1}{p} / 10^{-5} \text{ Pa}^{-1}$
100	2.00	0.5
200	1.00	1.0
300	0.67	1.5
400	0.50	2.0

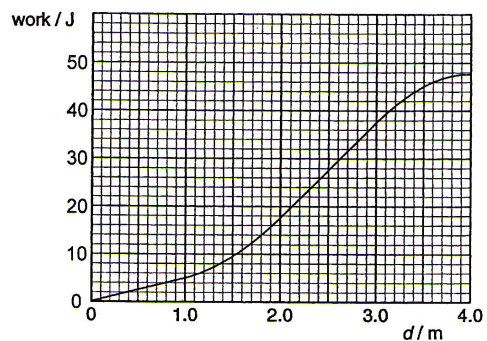


- (ii) The graph of $\frac{1}{p}$ against V is a straight line through the origin. This shows that $\frac{1}{p}$ is directly proportional to V and $pV = \text{constant}$, hence $p = \text{constant}/V$.

Note: Many failed to mention that the line must pass through the origin.

(b)

d / m	Work done / J
1.0	5.0
2.0	17.5
3.0	37.5
4.0	47.5



Note: Free-hand sketches of straight lines should be discouraged.

(c) (i) Gradient = $\frac{-0.72 - 0.86}{7.5 - 6.7} = -\frac{1.58}{0.8} = -2.0$

(ii) The equation is of the form
 $\lg g = -2 \lg r + \lg c$ where $\lg c$ is the y-intercept of this graph.

$$\lg g = \lg r^{-2} + \lg c$$

$$\lg g = \lg cr^{-2}$$

$$g = cr^{-2}$$

The gradient of the graph shows that the gravitational field strength g varies according to the inverse square law with distance r from the centre of the Earth.

Note: A common starting point was to assume the relation between g and r and to then take logarithms of both sides of the equation, ignoring the answer in (c)(i).

2. **(c) (i)** When the coil rotates, there is a rate of change magnetic flux linking to the circuit